

Cone Covering Under Curvature Bounds at the Admissible Scale

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Source Question and Status

Amenta and Kemppainen prove a cone-covering lemma for complete, geometrically doubling Riemannian manifolds of nonnegative sectional curvature. They then write that they know no doubling Riemannian manifold on which it fails and ask for a more precise determination of the spaces admitting such coverings [1, Remark B.6].

Remark B.6. At the time of writing we do not know of any doubling Riemannian manifolds (equipped with ϕ and m) for which the cone covering lemma fails. It would be interesting to determine more precisely which spaces admit cone coverings of the type above.

This packet gives a broad sufficient condition allowing negative curvature: sectional curvature may be as negative as a fixed multiple of $-m(x)^{-2}$ on the ball of radius comparable to the admissibility scale $m(x)$. It includes the globally lower-bounded-curvature case whenever m is bounded, and therefore every compact Riemannian manifold with a bounded admissibility function satisfying the source assumptions. Status: *substantial partial result, likely valid*, subject to human review.

Setup and Proof Intuition

Retain the source's complete geometrically doubling Riemannian manifold (X, g) , weighted measure γ , and admissibility function $m : X \rightarrow (0, \infty)$. Thus γ is doubling on every fixed enlargement of the admissible balls and, for every $a > 0$, there is c_a such that

$$d(x, y) \leq am(x) \implies m(x) \leq c_a m(y). \quad (1)$$

Put $\beta = c_1$, as in the proof of source Lemma B.4, and fix the extension E^* supplied by source Lemma B.3 for this β .

Nonnegative curvature enters that proof only once: it supplies a finite net of radial directions whose geodesics remain within $s/4$ of every other radial geodesic. Negative curvature makes radial geodesics separate, but Rauch comparison quantifies the separation. If curvature is controlled on the scale $m(x)$, a uniformly finer angular net restores the same $s/4$ estimate. The measure-theoretic part of the source proof is then unchanged.

Admissible-Scale Angular Comparison

Lemma 1. *Suppose that for some $\kappa \geq 0$ and a point $x \in X$,*

$$\sec_z \geq -\frac{\kappa^2}{m(x)^2} \quad \text{for every } z \in \overline{B}(x, \beta m(x)). \quad (2)$$

Set

$$\delta_{\kappa,\beta} = \begin{cases} \frac{\kappa\beta}{4 \sinh(\kappa\beta)}, & \kappa > 0, \\ \frac{1}{4}, & \kappa = 0. \end{cases} \quad (3)$$

If $v, w \in T_x X$ are unit vectors with $\angle(v, w) \leq \delta_{\kappa,\beta}$, then their unit-speed geodesics satisfy

$$d(\gamma_v(s), \gamma_w(s)) \leq s/4 \quad (0 \leq s \leq \beta m(x)).$$

Proof. Join v to w by the short great-circle arc in the unit tangent sphere and vary the corresponding radial geodesics. The variation field is an orthogonal Jacobi field J with $J(0) = 0$ and unit initial derivative when the angular parameter is arclength. Rauch comparison under (2) gives

$$|J(s)| \leq \begin{cases} \frac{m(x)}{\kappa} \sinh\left(\frac{\kappa s}{m(x)}\right), & \kappa > 0, \\ s, & \kappa = 0. \end{cases}$$

The distance between the two endpoints is at most the length of this angular variation. For $q = \kappa s/m(x) \leq \kappa\beta$ and $\kappa > 0$,

$$\frac{d(\gamma_v(s), \gamma_w(s))}{s} \leq \angle(v, w) \frac{\sinh q}{q} \leq \frac{1}{4} \frac{\kappa\beta}{\sinh(\kappa\beta)} \frac{\sinh q}{q} \leq \frac{1}{4},$$

because $q \mapsto \sinh(q)/q$ is increasing. The limiting case $\kappa = 0$ is immediate. $\square$

Main Partial Result

Theorem 1 (cone covering at the curvature scale). *Assume the source hypotheses above and suppose that there is a fixed $\kappa \geq 0$ such that, for every $x \in X$,*

$$\sec_z \geq -\frac{\kappa^2}{m(x)^2} \quad (z \in \overline{B}(x, \beta m(x))). \quad (4)$$

Then the cone-covering conclusion of source Lemma B.4 holds: for every bounded open $E \subset X$ and $x \in E$, there are $x_1, \dots, x_N \in X \setminus E$, with N depending only on $\dim X$, β , and κ , such that

$$\Gamma(x) \setminus T(E^*) \subset \bigcup_{j=1}^N \Gamma(x_j).$$

Consequently, the source's pointwise estimate and its vector-valued tent-space applications hold under (4).

Proof. Choose a $\delta_{\kappa,\beta}$ -net $v_1, \dots, v_N$ in the unit sphere of $T_x X$. Compactness of the Euclidean unit sphere gives a cardinality bound depending only on $\dim X$, β , and κ . By Lemma 1, every radial unit geodesic γ from x has an index j such that

$$d(\gamma(s), \gamma_{v_j}(s)) \leq s/4 \quad (0 \leq s \leq \beta m(x)). \quad (5)$$

If $E = X$, then the left-hand side of the claimed covering is empty and the empty family suffices. Otherwise, for each j , let $t_j > 0$ be the first time at which $\overline{B}(\gamma_{v_j}(t_j), t_j/4)$ meets $X \setminus E$, and choose

$$x_j \in (X \setminus E) \cap \overline{B}(\gamma_{v_j}(t_j), t_j/4).$$

These times exist because E is bounded. Let $(y, t) \in \Gamma(x) \setminus T(E^*)$, write $r = d(x, y) < t$, and choose a minimizing unit geodesic γ from x to y . Pick j satisfying (5).

First suppose $t_j \leq \beta m(x)$. If $r > t_j$, then (5) gives

$$d(y, x_j) \leq (r - t_j) + t_j/4 + t_j/4 < r < t.$$

If $r \leq t_j$, then (5) shows that y belongs to the sector

$$R(v_j, t_j) = \bigcup_{0 \leq s \leq t_j} B(\gamma_{v_j}(s), s/4),$$

which is contained in E by minimality of t_j . Source Lemma B.3 gives $B(y, 2t_j) \subset E^*$. Since $(y, t) \notin T(E^*)$ means $B(y, t) \not\subset E^*$, one has $2t_j < t$. The triangle estimate from the source proof gives $d(y, x_j) \leq 2t_j < t$.

It remains to rule out $t_j > \beta m(x)$. Since $r < t < m(y)$, applying (1) with the roles of x, y reversed gives $t < m(y) \leq c_1 m(x) = \beta m(x)$. Equation (5) puts $y \in R(v_j, \beta m(x)) \subset E$. Source Lemma B.3 therefore gives $B(y, 2\beta m(x)) \subset E^*$, while $B(y, t) \not\subset E^*$. This forces $\beta m(x) < t$, a contradiction. Thus one of the selected x_j satisfies $d(y, x_j) < t$, proving the desired cover. $\square$

Corollary 1. *Suppose $\sec X \geq -K$ globally and $m(x) \leq M$ for all x . Then the cone-covering property holds, with constants depending additionally on $\sqrt{KM}$. In particular, it holds on every compact Riemannian manifold for every bounded admissibility function satisfying the source's local doubling and comparability assumptions.*

Proof. Take $\kappa = \sqrt{KM}$. Since $m(x) \leq M$, one has $-K \geq -\kappa^2/m(x)^2$, so (4) follows from the global lower curvature bound. A compact Riemannian manifold has sectional curvature bounded below. $\square$

Scope, Upgrade Attempts, and Verification

The theorem is a sufficient condition, not a characterization. It rules out compact manifolds and the usual bounded-admissibility, lower-curvature settings as sources of counterexamples. It also allows unbounded negative curvature when $m(x)$ shrinks at least at the reciprocal curvature scale. Rauch comparison genuinely uses sectional, rather than merely Ricci, curvature.

An additional attempt to use geometric doubling alone did not close the problem. Doubling controls endpoint packings at each fixed scale, whereas the proof requires a coherent radial net at all smaller scales; no such implication or counterexample was found. Hyperbolic space illustrates the radial obstruction but is not doubling.

The run's cheap indexes contained no answer. Searches through 17 August 2026 found the source and the later abstract use of the cone-covering property in [3], but no counterexample or this admissible-scale negative-curvature extension. The companion script samples the exact dimensionless inequality behind (3); it is a consistency check, not a substitute for Rauch comparison.

Human Review Recommendation

Verify that the Rauch estimate is applicable to the full angular variation inside $B(x, \beta m(x))$, including beyond the cut locus, and compare the two case splits after (5) with source Lemma B.4. These are the only new geometric and bookkeeping steps.

References

- [1] A. Amenta and M. Kemppainen, *Non-uniformly local tent spaces*, Publ. Mat. **59** (2015), 245–270; arXiv:1309.5758.
- [2] M. do Carmo, *Riemannian Geometry*, Birkhäuser, 1992; chapter on the Rauch comparison theorem.
- [3] M. Kemppainen, *On vector-valued tent spaces and Hardy spaces associated with non-negative self-adjoint operators*, Glasgow Math. J. **58** (2016), 689–716; arXiv:1402.2886.